

Business Intelligence & Analytics Project

E-commerce Analytics and Customer Insights

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0.1 Problem Statement

The rapid growth of e-commerce has generated massive volumes of transactional and customer behavioral data. Businesses require analytical frameworks to transform raw data into actionable insights that improve customer satisfaction, retention, operational efficiency, and profitability. This project investigates customer purchasing behavior and operational performance through business intelligence techniques using Python for pre-processing and analytics, and Power BI for dashboard development.

The objective of this project is to:

- Analyze retail transactions
- Monitor KPIs
- Identify customer purchasing patterns
- Forecast future sales
- Support strategic business decisions

0.2 Dataset Description

The **Online Retail II** dataset is a prominent dataset within the UCI Machine Learning Repository, widely used for business intelligence, customer behavior analysis, and predictive modeling.

Source and Context

The data was originally collected from a **UK-based, non-store online retailer** that specializes in unique all-occasion giftware. The transactions occurred between **2009 and 2011**, providing a comprehensive view of business performance across two full years.

Dataset Structure

The dataset consists of approximately **525,461 rows** for the 2009–2010 period. It is structured around the following key columns which allow for detailed transactional analysis:

- **Invoice:** A unique 6-digit integral number assigned to each transaction; codes starting with 'C' indicate cancellations.
- **StockCode:** A unique 5-digit integral number assigned to each distinct product.
- **Description:** The nominal name of the product.
- **Quantity:** The quantities of each product per transaction.
- **InvoiceDate:** The day and time when a transaction was generated.
- **Price:** The unit price of the product in sterling (£).

- **Customer ID:** A unique 5-digit integral number assigned to each customer.
- **Country:** The name of the country where the customer resides.

Chapter 1

PowerBI

1.1 Data Preparation and Modeling Process

The dataset was prepared and modeled in Power BI through a structured workflow involving data import, cleaning, transformation, and modeling. The objective was to create a reliable analytical model that supports dashboard development and business insights.

1.1.1 Data Import and Integration

Two separate datasets covering the periods 2009–2010 and 2010–2011 were imported into Power BI. These datasets were reviewed to ensure consistency in structure and formatting before integration.

After initial inspection, both datasets were appended into a single consolidated table named **FactSales**, forming the core transactional dataset for the analysis.

1.1.2 Data Cleaning (Power Query)

Data cleaning was performed using Power Query Editor to ensure accuracy, consistency, and reliability of the dataset. The following cleaning operations were applied:

- Removed records with null Customer IDs to maintain valid customer-level analysis
- Excluded cancelled invoices identified by invoice numbers starting with “C”
- Eliminated records with negative quantities and invalid price values
- Removed rows with missing or inconsistent product descriptions
- Standardized the dataset to ensure uniform formatting across both years

This step ensured that only valid and meaningful transactional data was retained for analysis.

1.1.3 Feature Engineering

To enhance analytical capabilities, additional calculated fields were created:

- A Revenue column was generated using:

$$\text{Revenue} = \text{Quantity} \times \text{Price} \tag{1.1}$$

- Multiple date-related attributes were extracted from the invoice date, including:
 - Year
 - Month
 - Month Name
 - Quarter
 - Day Name
 - Hour

These features enable detailed time-based analysis such as trends, seasonality, and peak activity periods.

1.1.4 Data Modeling (Star Schema)

A star schema model was designed to structure the dataset efficiently for analysis and visualization. The final model consists of:

- **Fact Table:** FactSales (merged and cleaned transactional data)
- **Dimension Tables:**
 - DimCustomer
 - DimProduct
 - DimDate

The FactSales table serves as the central table, while dimension tables provide descriptive context for analysis.

1.1.5 Relationships

Relationships were established between the FactSales table and all dimension tables using primary and foreign key relationships.

Each dimension table is connected to FactSales in a one-to-many relationship, ensuring:

- Data integrity
- Efficient filtering
- Accurate aggregation across visuals
- A scalable and optimized data model

1.2 DAX Measures

After structuring the model, key DAX measures were created to support KPIs and dashboard analytics. These measures were designed for reusability across all reports.

1.2.1 Total Revenue

`Total Revenue = SUM(FactSales[Revenue])`

This measure calculates the total revenue generated from all valid transactions in the FactSales table.

1.2.2 Total Orders

`Total Orders = DISTINCTCOUNT(FactSales[Invoice])`

This measure counts the number of unique invoices, representing the total number of orders placed.

1.2.3 Total Customers

`Total Customers = DISTINCTCOUNT(FactSales[Customer ID])`

This measure calculates the number of unique customers in the dataset.

1.2.4 Total Products Sold

`Total Products Sold = SUM(FactSales[Quantity])`

This measure represents the total quantity of products sold across all transactions.

1.2.5 Average Order Value

`Average Order Value = DIVIDE([Total Revenue], [Total Orders])`

This KPI measures the average revenue generated per order, providing insight into customer spending behavior.

1.3 Model Readiness for Visualization

After completing data cleaning, transformation, and modeling, the dataset was fully optimized and structured for dashboard development.

The final model is:

- Clean and validated
- Fully relational (star schema)
- Enriched with calculated fields
- Equipped with reusable DAX measures

This ensures that dashboards can now be built efficiently without requiring further backend modifications, enabling a smooth transition to the visualization and insight phase.

1.4 Dashboard Development

1.4.1 Executive Overview Dashboard

The Executive Overview dashboard focuses on providing high-level business insights and overall sales performance analysis. The dashboard was designed to help management monitor revenue performance, sales growth, customer activity, and country-level sales distribution.

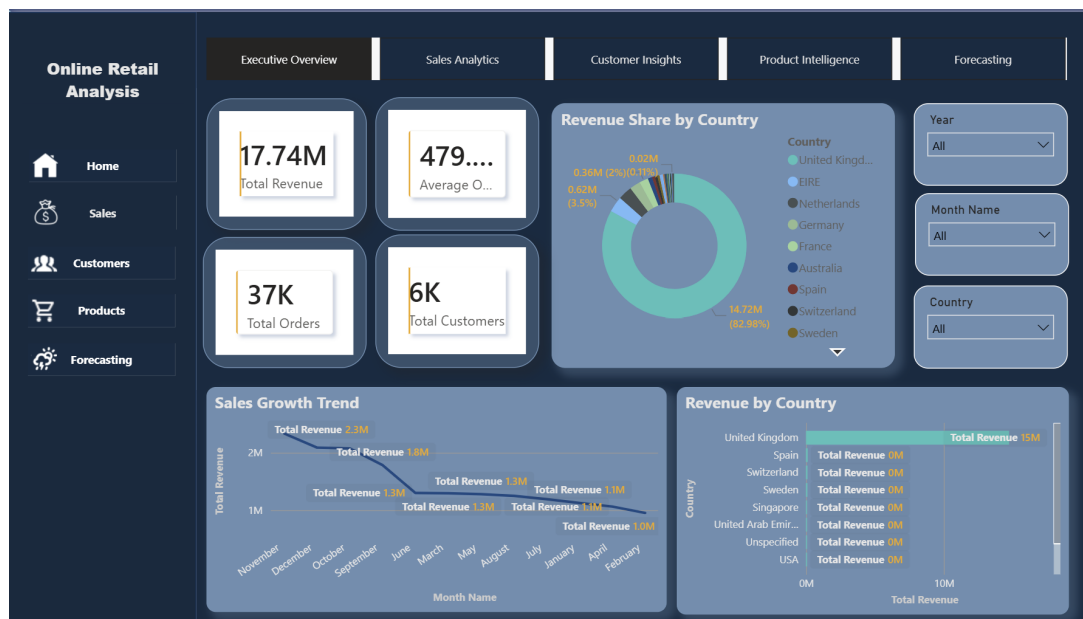


Figure 1.1: Executive Overview Dashboard

This dashboard includes:

Several KPI cards and visualizations were implemented to provide a complete overview of business performance metrics.

KPI Cards

Total Revenue: A KPI card was created using the **Total Revenue** measure to display the total revenue generated from all sales transactions. **Purpose:**

- Measure overall business performance
- Monitor revenue growth over time

Total Orders: A KPI card was implemented using the **Total Orders** measure to display the total number of completed orders.

Purpose:

- Measure sales activity
- Analyze order volume trends

Total Customers: A KPI card was created using the **Total Customers** measure to display the total number of unique customers within the dataset.

Purpose:

- Analyze customer reach
- Track customer base size

Average Order Value: A KPI card was implemented using the **Average Order Value** measure to calculate the average revenue generated per order.

Purpose:

- Analyze customer purchasing behavior
- Evaluate average transaction value

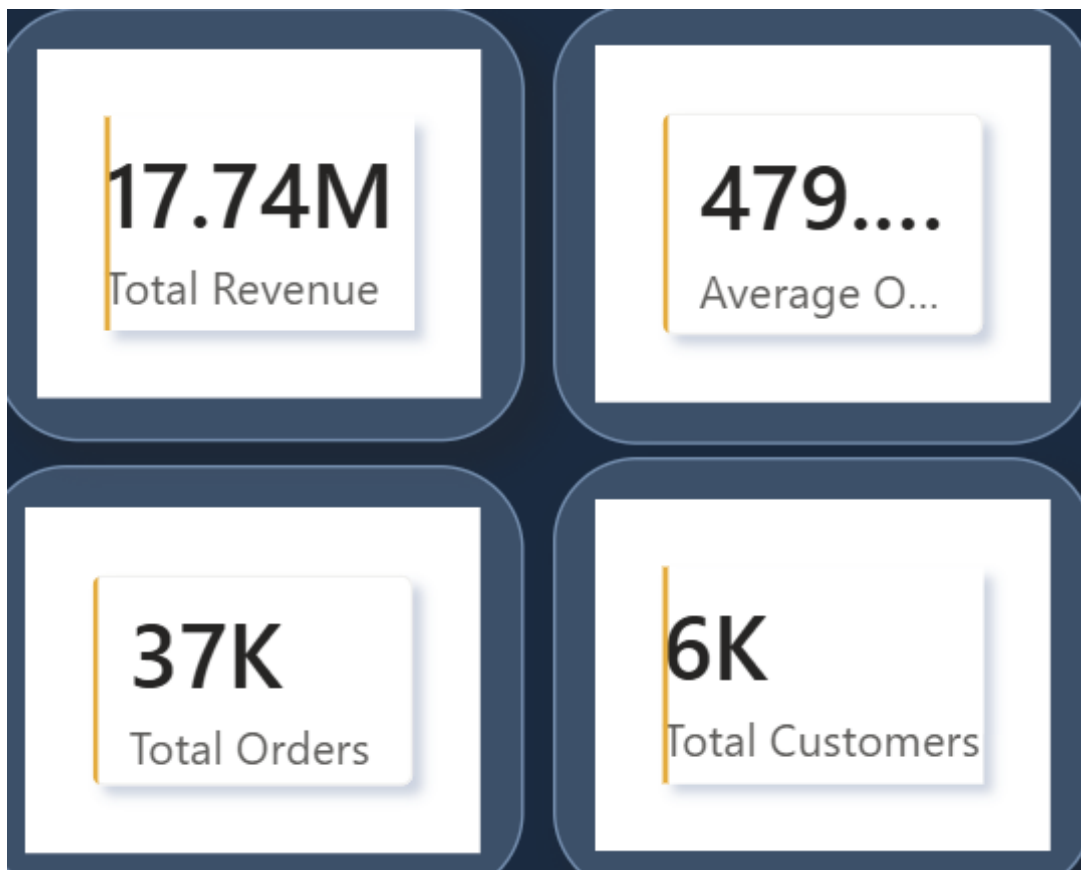


Figure 1.2: Executive Dashboard KPI Cards

Visualizations

Revenue Trend Over Time: A **Line Chart** was implemented to visualize revenue trends across different months.

Configuration:

- X-Axis: Month Name

- Y-Axis: Total Revenue

Purpose:

- Analyze sales growth trends
- Identify seasonal sales patterns
- Monitor monthly business performance

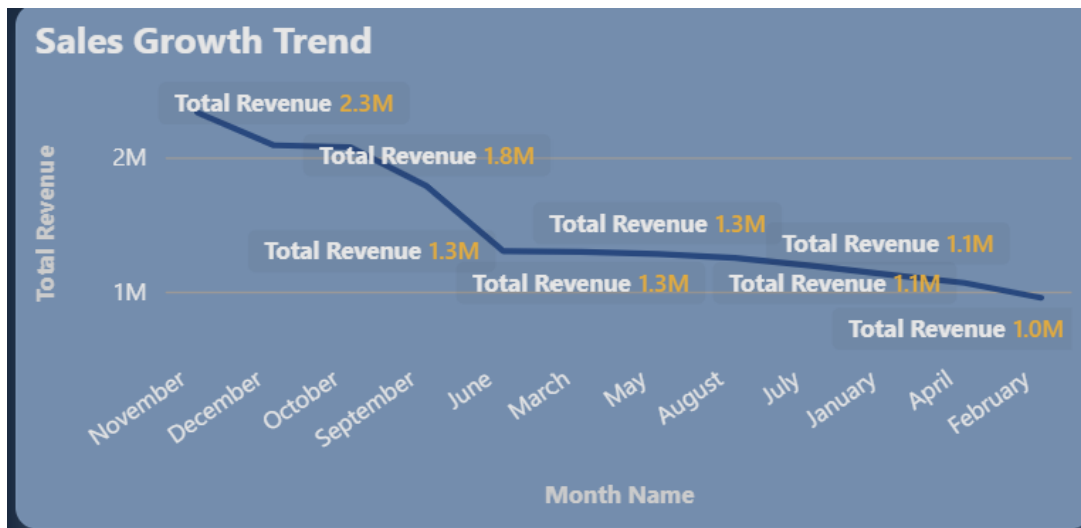


Figure 1.3: Revenue Trend Over Time

Top Revenue-Generating Countries: A **Clustered Bar Chart** was created to display countries generating the highest revenue.

Configuration:

- Y-Axis: Country
- X-Axis: Total Revenue

Purpose:

- Identify strongest business markets
- Compare sales performance between countries

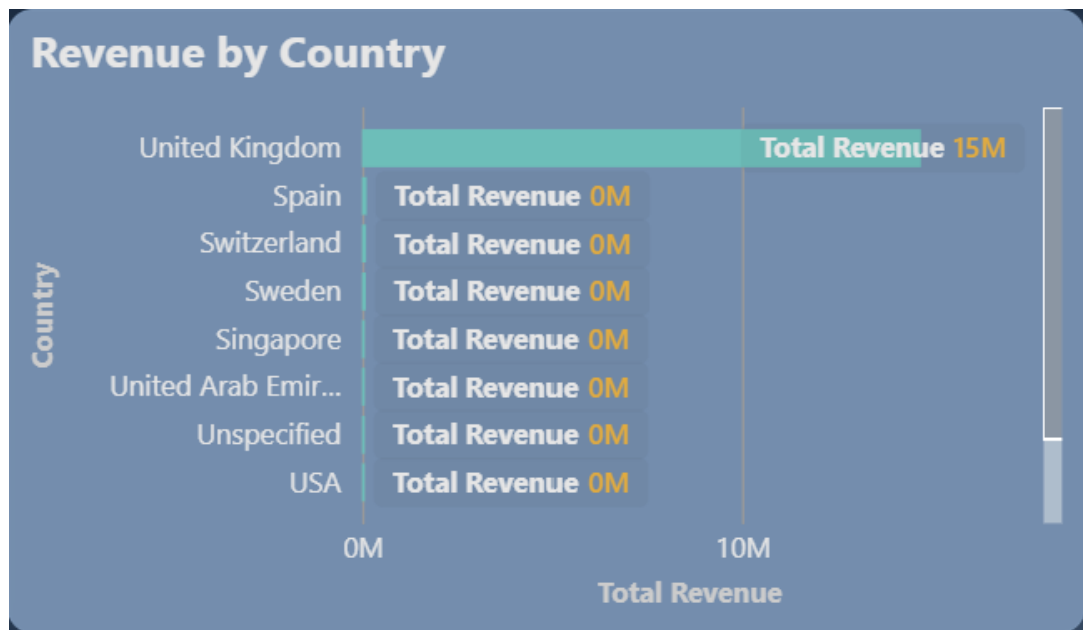


Figure 1.4: Top Revenue-Generating Countries

Revenue Share by Country: A **Donut Chart** was implemented to visualize the contribution of different countries to total revenue.

Configuration:

- Legend: Country
- Values: Total Revenue

Purpose:

- Analyze geographic revenue distribution
- Identify dominant business markets

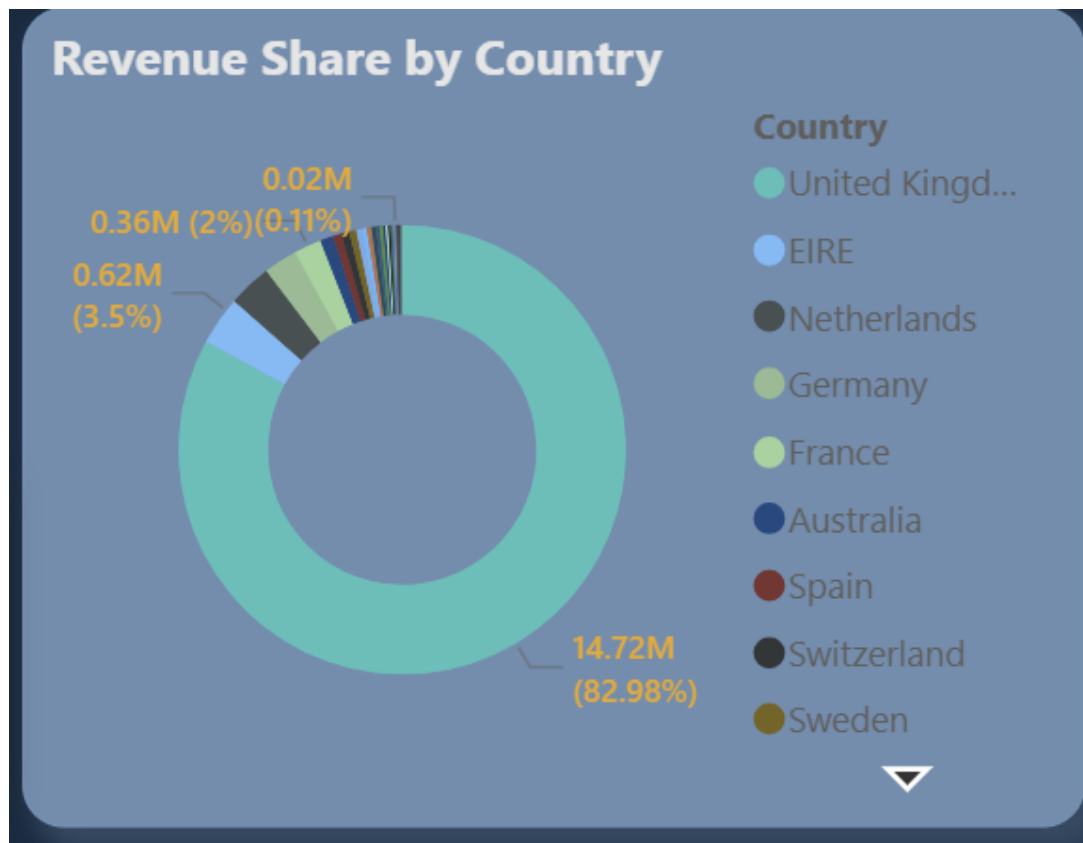


Figure 1.5: Revenue Share by Country

Interactive Slicers: Several slicers were implemented to improve dashboard interactivity and allow dynamic filtering of business insights.

Configuration:

- Year
- Month Name
- Country

Purpose:

- Enable dynamic report filtering
- Improve user interaction and dashboard navigation
- Support detailed sales analysis across different periods and countries

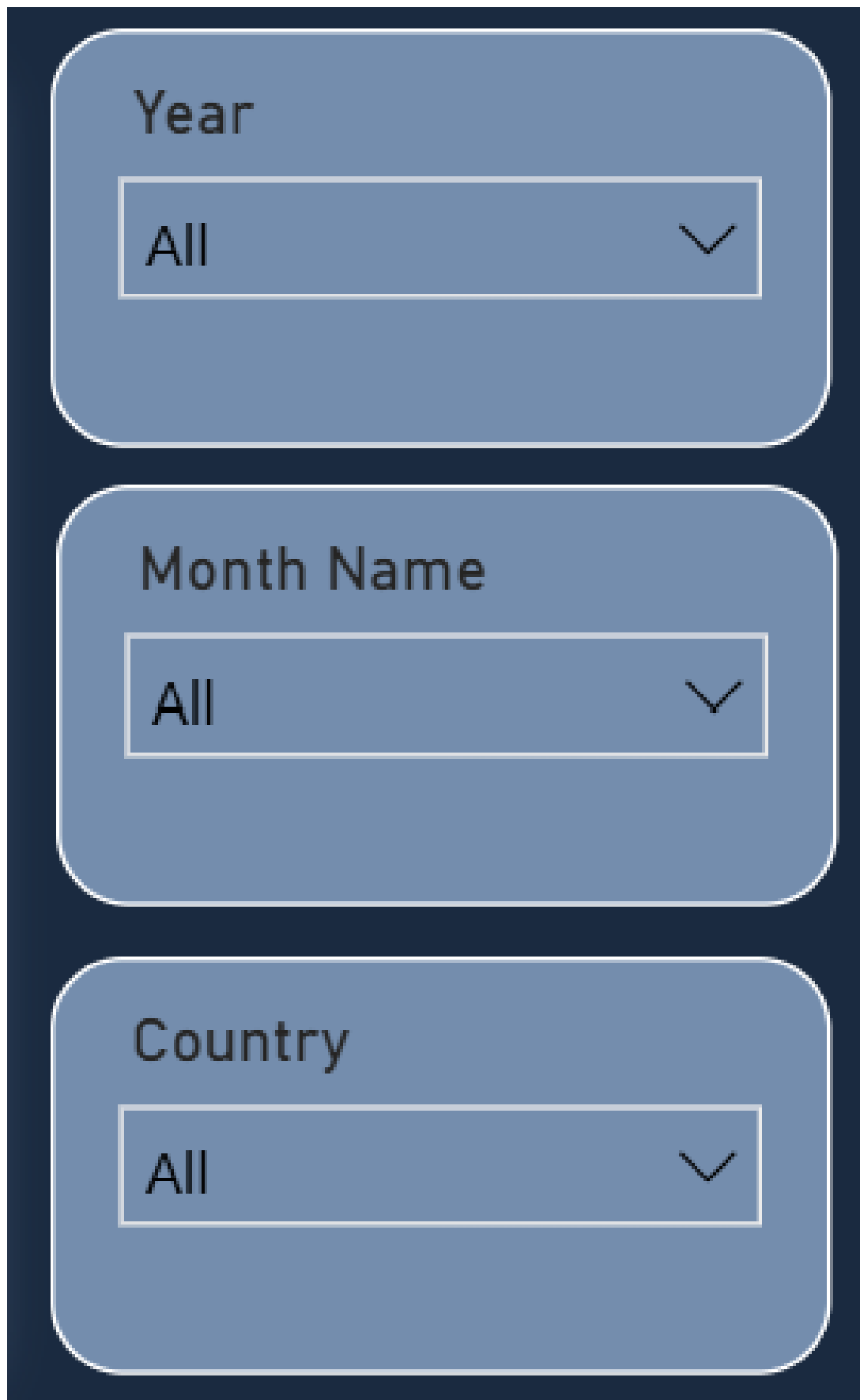


Figure 1.6: Executive Dashboard Slicers

1.4.2 Sales Analytics Dashboard

The Sales Analytics dashboard focuses on analyzing detailed sales behavior and identifying operational sales patterns.

The dashboard helps businesses understand purchasing trends, product performance, and customer shopping activity.

Several visualizations were implemented to provide deeper sales analysis and operational insights.

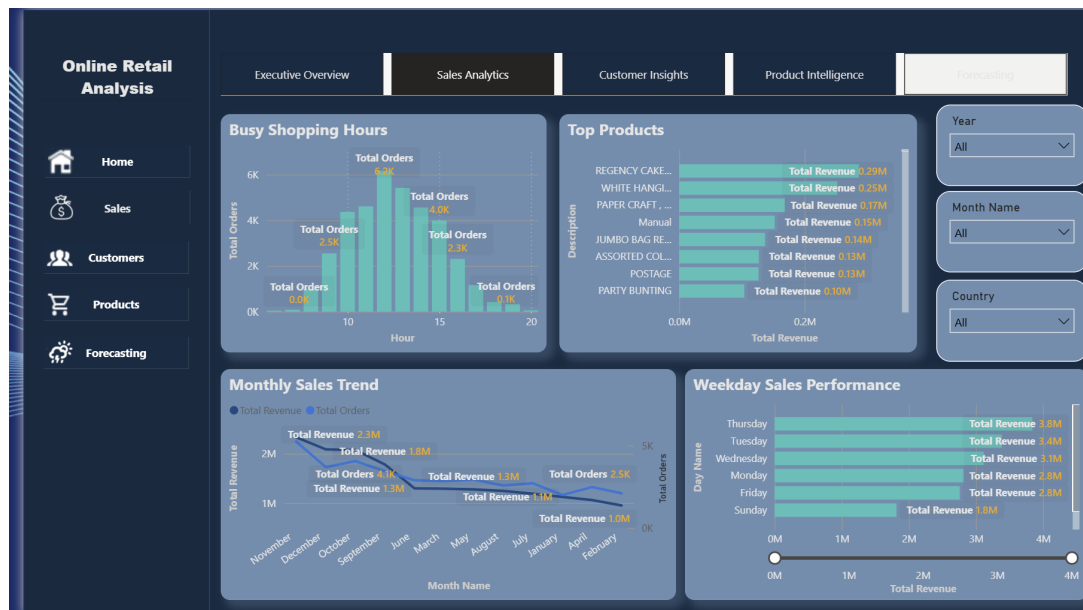


Figure 1.7: Sales Analytics Dashboard

This dashboard includes:

Visualizations

Monthly Sales Performance: A **Line Chart** was implemented to visualize monthly sales performance trends. **Configuration:**

- X-Axis: Month Name
- Y-Axis: Total Revenue

Purpose:

- Analyze monthly sales fluctuations
- Identify seasonal demand patterns

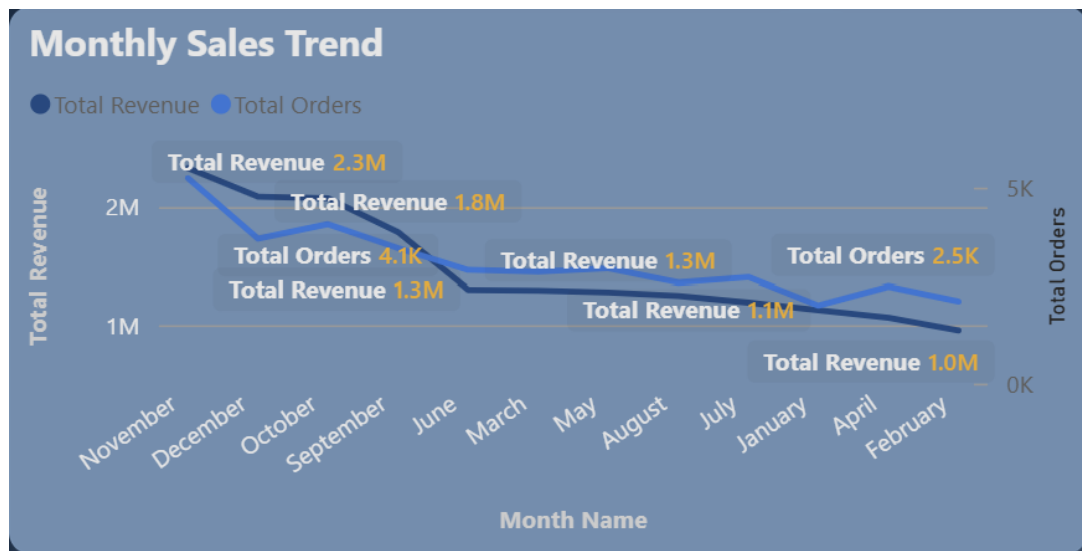


Figure 1.8: Monthly Sales Performance

Top Performing Products: A **Clustered Bar Chart** was created to display the top-performing products based on generated revenue.

Configuration:

- Y-Axis 1: Total Revenue
- Y-Axis 2: Total Orders
- X-Axis: Months Name

Purpose:

- Identify products generating the highest revenue
- Support inventory and sales strategy decisions



Figure 1.9: Top Performing Products

Peak Shopping Hours: A **Column Chart** was implemented to analyze customer purchasing activity across different hours of the day.

Configuration:

- X-Axis: Hour
- Y-Axis: Total Orders

Purpose:

- Identify busiest shopping periods
- Analyze customer purchasing behavior



Figure 1.10: Peak Shopping Hours

Sales Performance by Weekday: A **Bar Chart** was implemented to visualize revenue performance across weekdays.

Configuration:

- Y-Axis: Day Name
- X-Axis: Total Revenue

Purpose:

- Analyze customer purchasing patterns throughout the week
- Identify strongest sales days

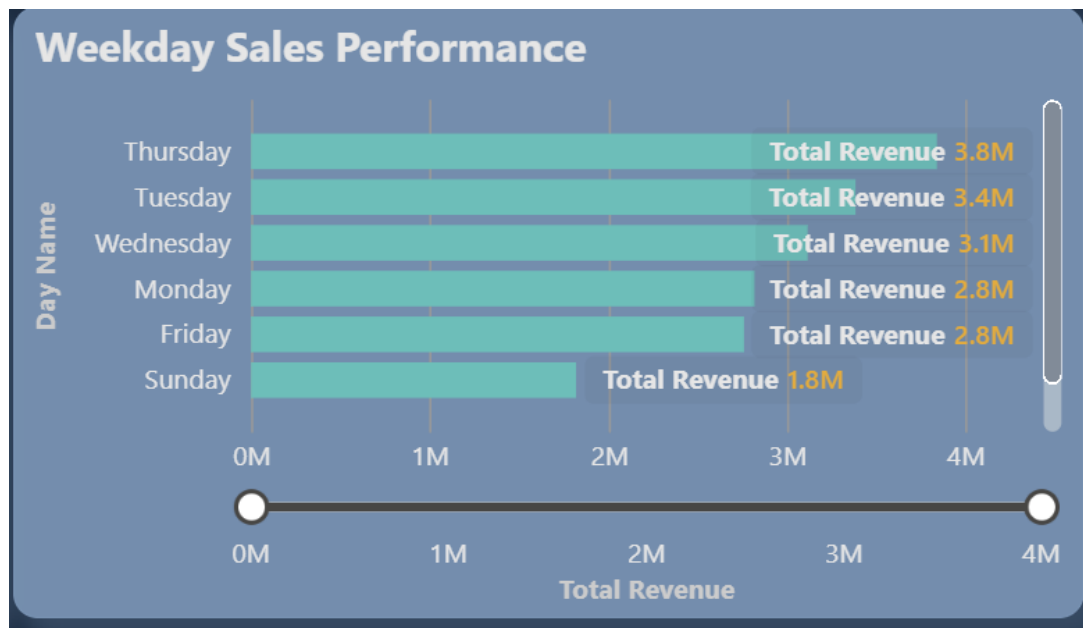


Figure 1.11: Sales Performance by Weekday

Interactive Slicers: Interactive slicers were added to support detailed sales analysis and improve report usability.

Configuration:

- Year
- Month Name
- Country

Purpose:

- Allow dynamic filtering of sales data
- Improve dashboard interactivity
- Support flexible analysis across multiple dimensions

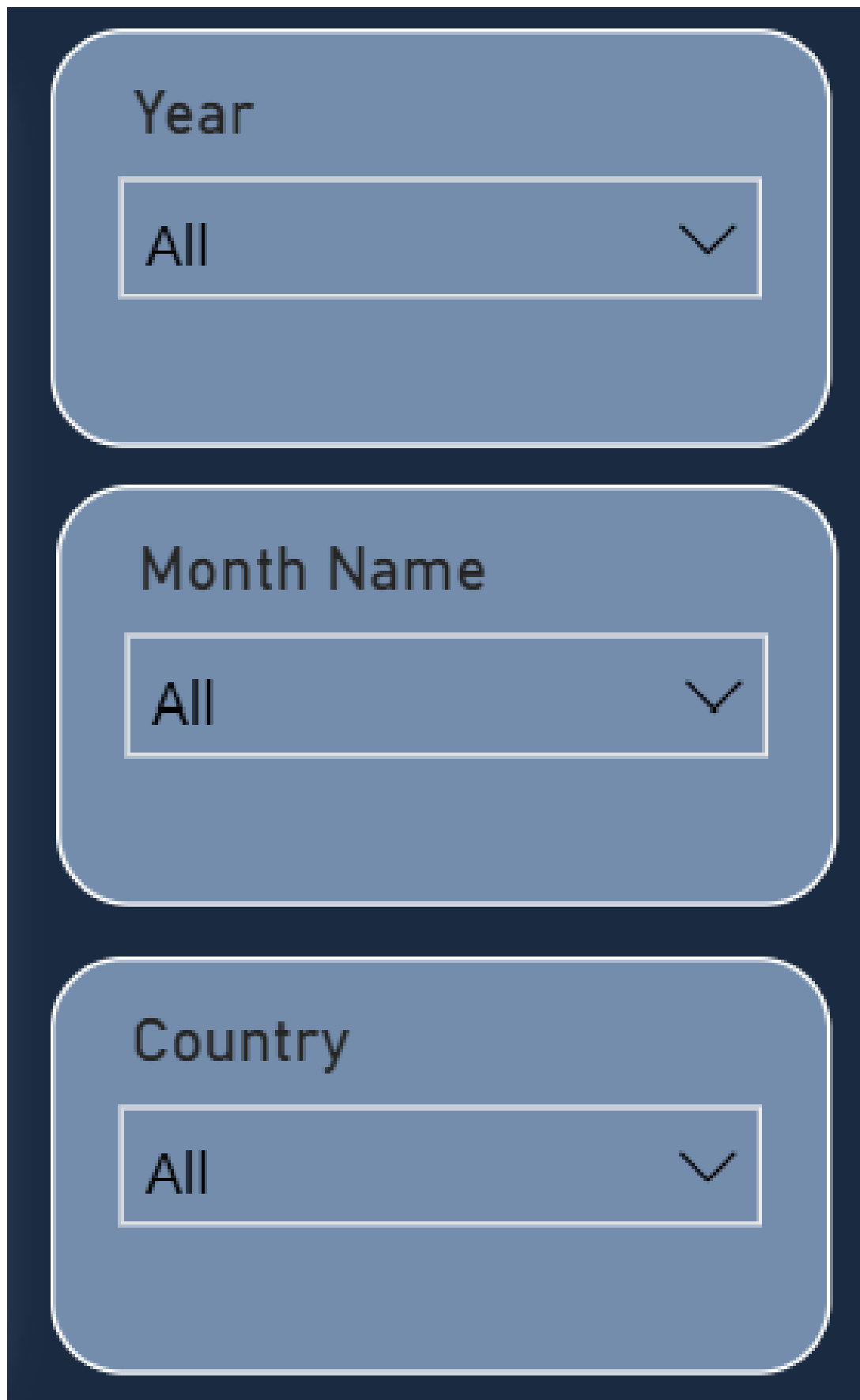


Figure 1.12: Sales Analytics Dashboard Slicers

1.4.3 Customer Insights Dashboard

- The Customer Insights dashboard focuses on understanding customer purchasing behavior, identifying valuable customers, and analyzing customer retention patterns.

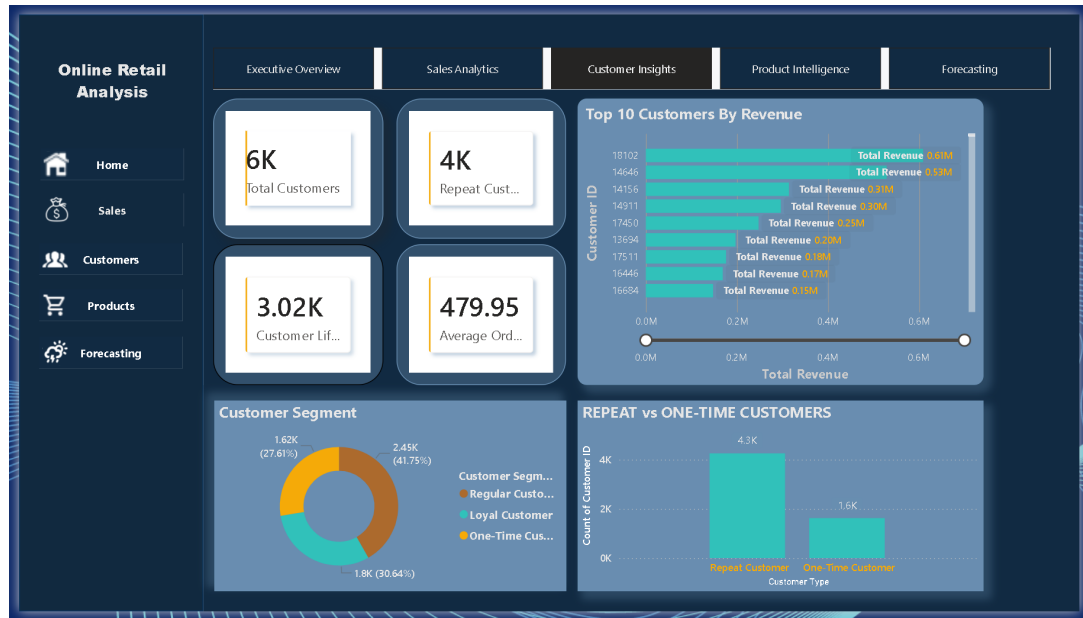


Figure 1.13: Customer Insights Dashboard

- Several KPI cards and visualizations were implemented to provide a complete overview of customer-related performance metrics.

KPI Cards

Total Customers: A KPI card was created using the **Total Customers** measure to display the total number of unique customers within the dataset.

Purpose:

- Measure overall customer base size
- Track customer growth over time

Repeat Customers: A KPI card was created using the **Repeat Customers** measure to display the number of customers who placed more than one order.

Purpose:

- Measure customer retention
- Identify loyal customers

DAX Measure:

```
Repeat Customers =
CALCULATE(
    DISTINCTCOUNT(FactSales[Customer ID]),
```

```

    FILTER(
        VALUES(FactSales[Customer ID]),
        CALCULATE(DISTINCTCOUNT(FactSales[Invoice])) > 1
    )
)

```

Customer Lifetime Value (CLV): A KPI card was implemented using the **Customer Lifetime Value** measure to calculate the average revenue generated per customer.

Purpose:

- Evaluate customer profitability
- Identify high-value customer groups

DAX Measure:

```

Customer Lifetime Value =
DIVIDE([Total Revenue], [Total Customers])

```

Average Order Value: A KPI card was created using the **Average Order Value** measure to calculate the average revenue generated per order.

Purpose:

- Measure customer spending behavior
- Analyze purchasing trends



Figure 1.14: KPI cards of Customer Insights Dashboard

Visualizations

Top 10 Customers by Revenue: A **Clustered Bar Chart** was implemented to display the top 10 customers generating the highest revenue.

Configuration:

- Y-Axis: Customer ID
- X-Axis: Total Revenue

Purpose:

- Identify top-performing customers
- Support customer relationship strategies

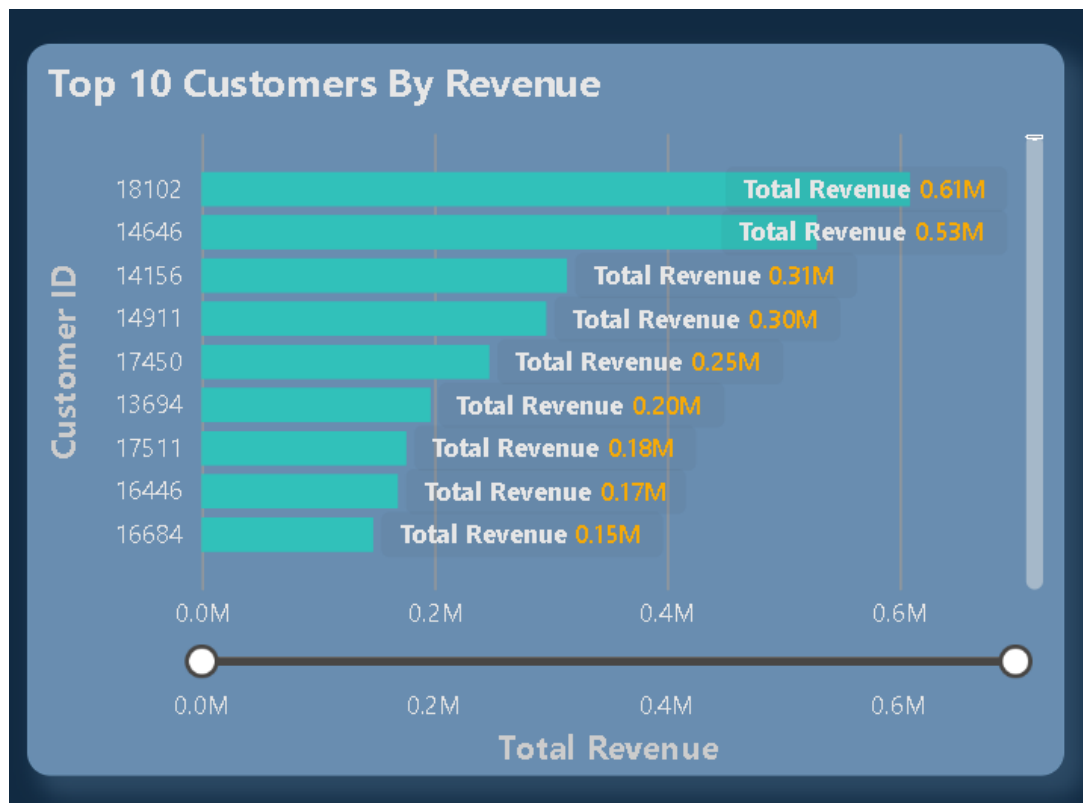


Figure 1.15: Top 10 Customers By Revenue

Customer Segment Distribution: A **Donut Chart** was created to visualize customer segmentation.

Configuration:

- Legend: Customer Segment
- Values: Count of Customer ID

The segmentation was based on customer purchasing frequency using the following calculated column:

```
Customer Segment =
VAR CustomerOrders =
    CALCULATE(
        DISTINCTCOUNT(FactSales[Invoice]),
        ALLEXCEPT(FactSales, FactSales[Customer ID])
    )
RETURN
SWITCH(
    TRUE(),
    CustomerOrders = 1, "One-Time Customer",
    CustomerOrders <= 5, "Regular Customer",
    "Loyal Customer"
)
```

Purpose:

- Understand customer loyalty categories
- Support targeted marketing strategies

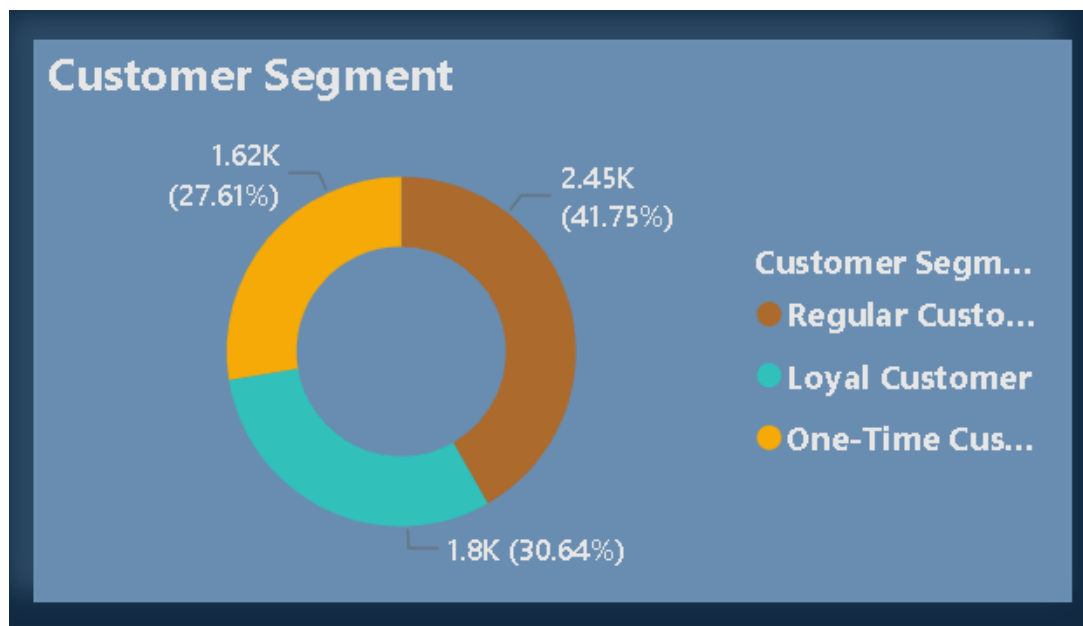


Figure 1.16: Customer Segment Donut Chart

Repeat vs One-Time Customers: A **Clustered Column Chart** was created to compare repeat customers and one-time customers.

Configuration:

- X-Axis: Customer Type
- Y-Axis: Count of Customer ID

Purpose:

- Analyze customer retention performance
- Compare customer purchasing frequency

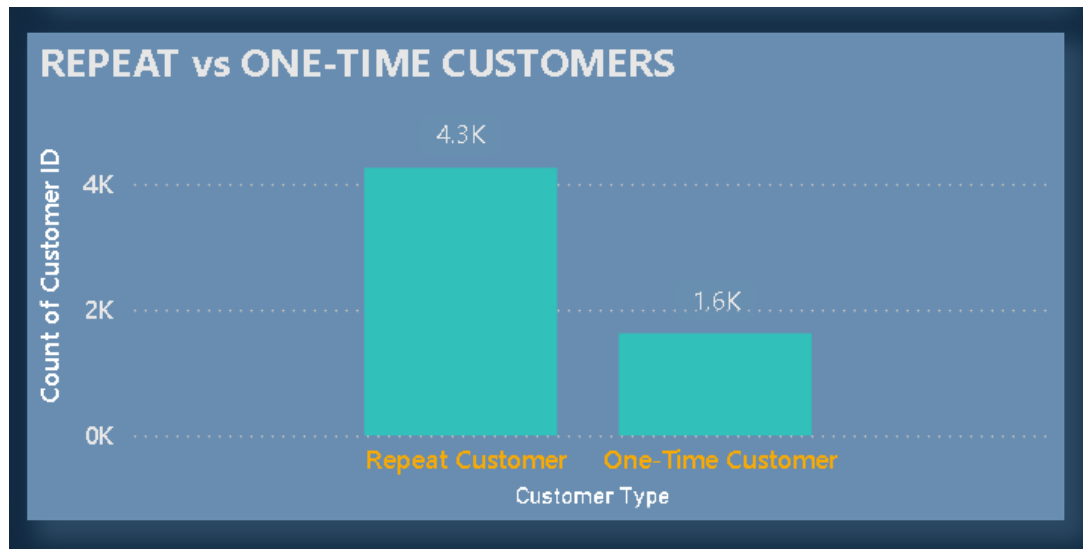


Figure 1.17: Repeat VS One-Time Customers

1.4.4 Product Intelligence Dashboard

The Product Intelligence dashboard focuses on evaluating product performance and identifying products contributing most to business revenue.

This dashboard helps businesses improve inventory management and sales strategies.

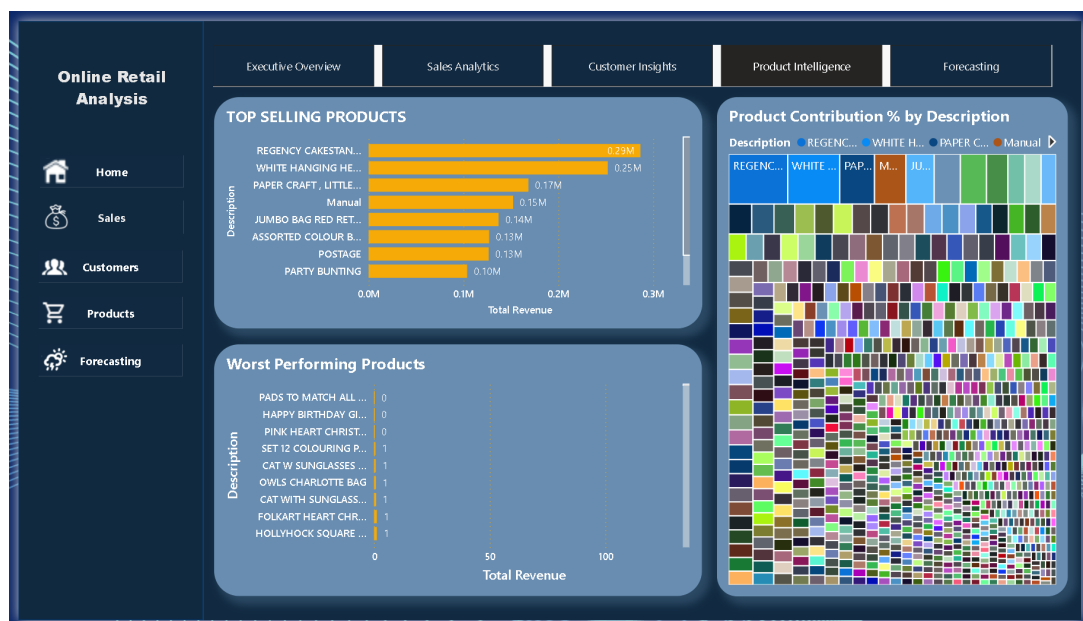


Figure 1.18: Product Intelligence Dashboard

Visualizations

Top Selling Products: A **Clustered Bar Chart** was implemented to display products generating the highest revenue.

Configuration:

- Y-Axis: Description

- X-Axis: Total Revenue

Purpose:

- Identify best-selling products
- Analyze high-performing inventory items

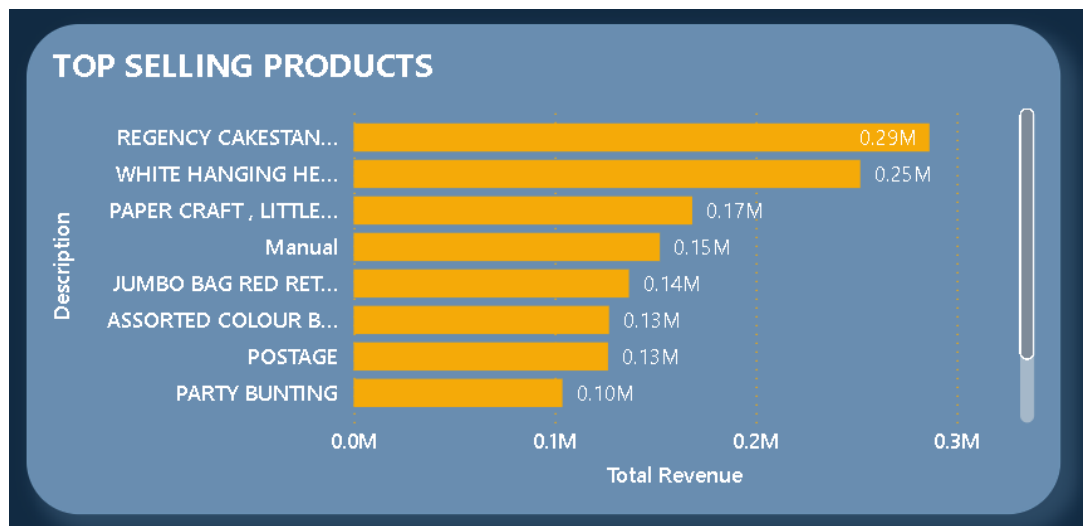


Figure 1.19: Top Selling Products Bar Chart

Worst Performing Products: A **Clustered Bar Chart** was created to identify products generating the lowest revenue.

Configuration:

- Y-Axis: Description
- X-Axis: Total Revenue
- Sorting: Ascending by Total Revenue

Purpose:

- Detect underperforming products
- Support inventory optimization decisions

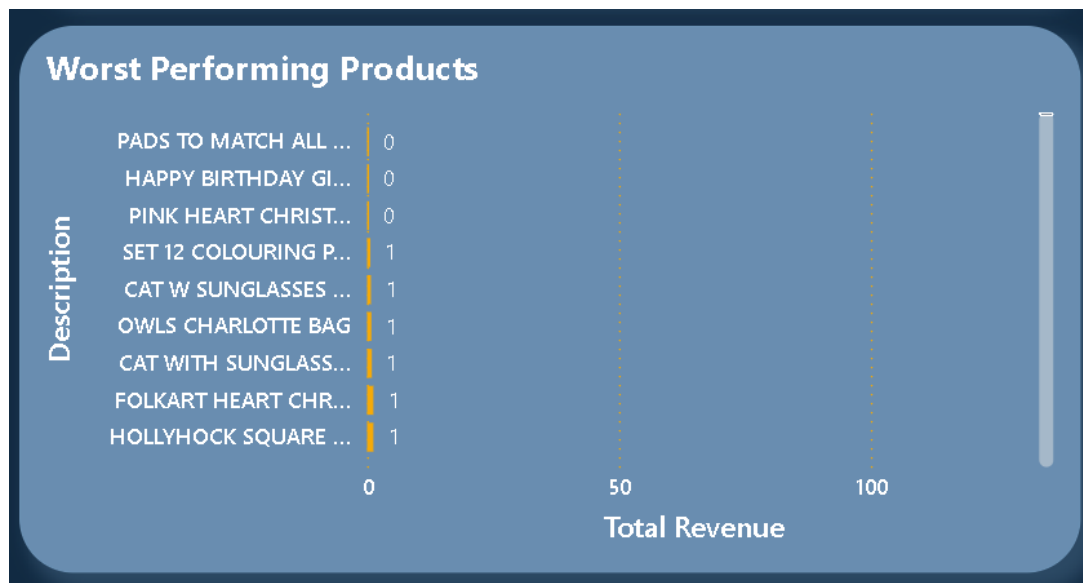


Figure 1.20: Worst Performing Products Bar Chart

Product Contribution Percentage by Description: A Treemap Visualization was implemented to show the percentage contribution of each product to total revenue.

Configuration:

- Category: Description
- Values: Product Contribution %

The measure was formatted as a percentage to improve readability.

DAX Measure:

```
Product Contribution % =
DIVIDE(
    [Total Revenue],
    CALCULATE(
        [Total Revenue],
        ALL(FactSales[Description])
    )
)
```

Purpose:

- Evaluate product contribution to overall business performance
- Identify revenue-driving products

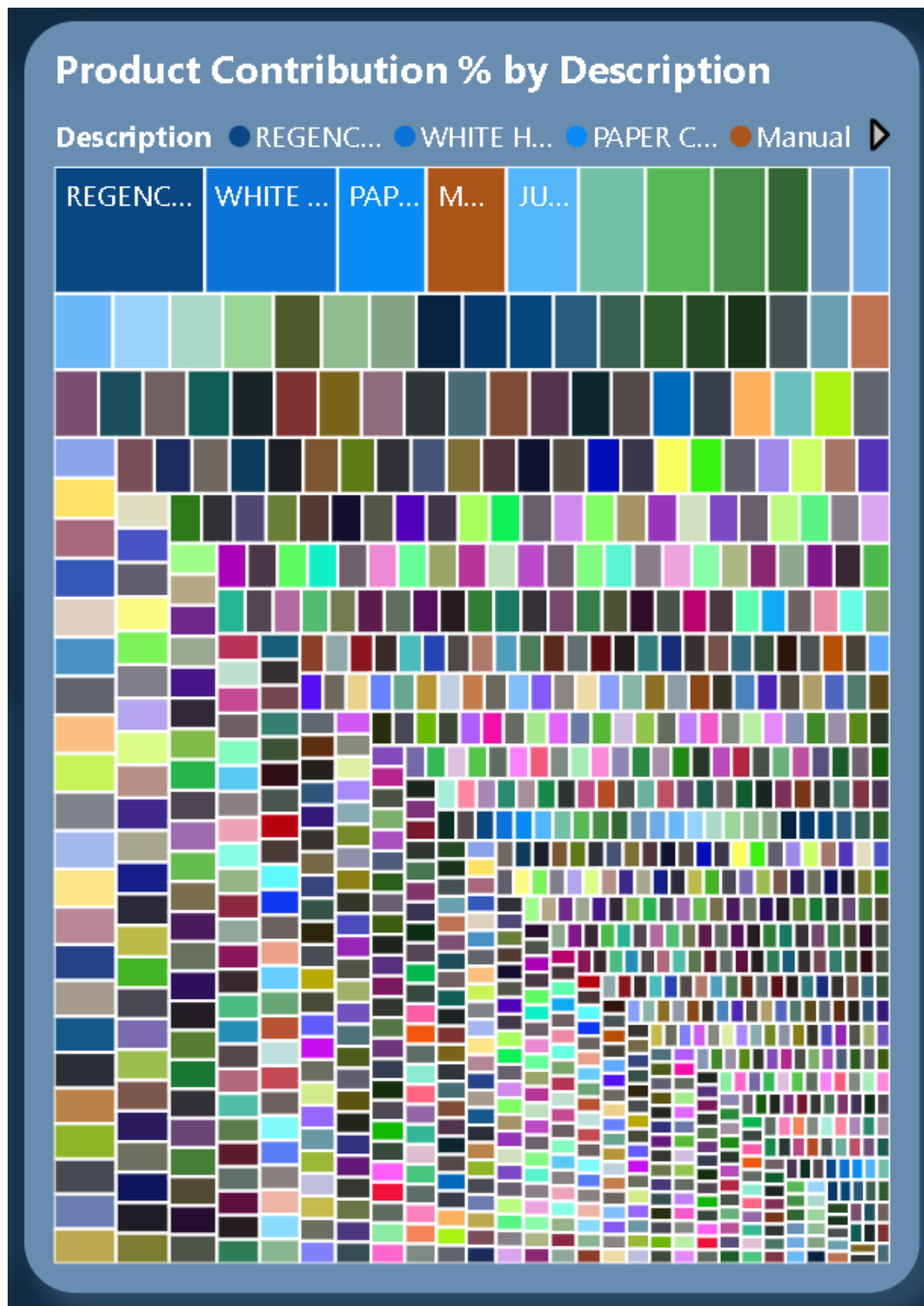


Figure 1.21: Product Contribution Percentage By Description

1.4.5 Forecasting Dashboard

The Forecasting dashboard was developed to support predictive analytics and future business planning.

Power BI forecasting tools were used to analyze historical revenue trends and estimate

future performance.

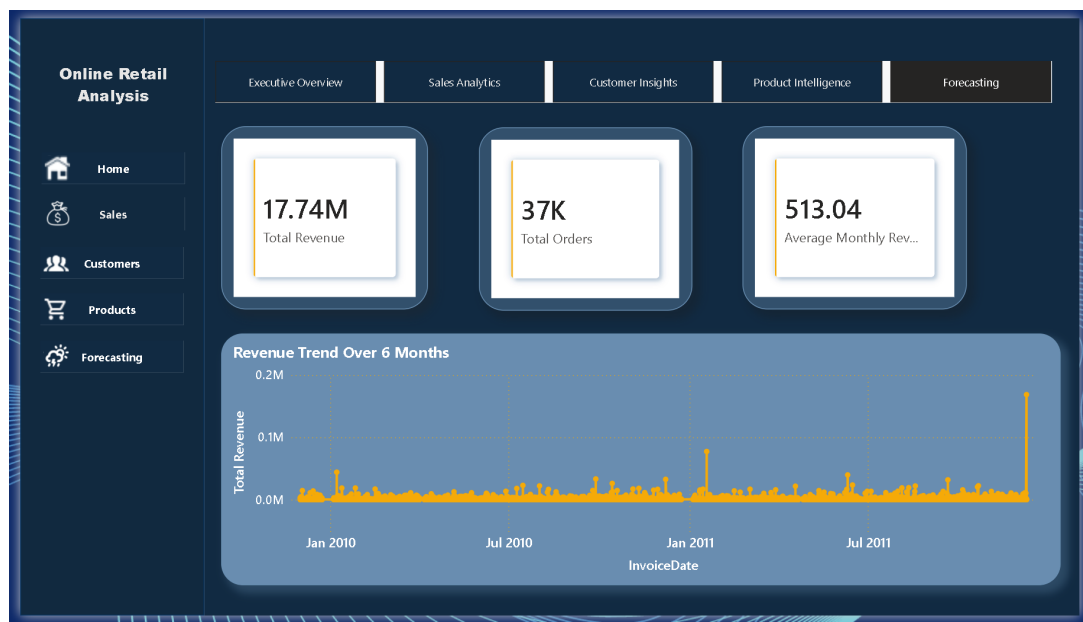


Figure 1.22: Forecasting Dashboard

KPI Cards

Total Revenue: A KPI card was implemented using the **Total Revenue** measure to display overall revenue generated by the business.

Purpose:

- Monitor financial performance
- Provide a high-level revenue overview

Total Orders: A KPI card was created using the **Total Orders** measure to display the total number of orders processed.

Purpose:

- Measure business activity
- Track operational performance

Average Monthly Revenue: A KPI card was implemented using the **Average Monthly Revenue** measure to calculate average revenue across invoice dates.

DAX Measure:

```
Average Monthly Revenue =  
AVERAGEX(  
    VALUES(FactSales[InvoiceDate]),  
    [Total Revenue]  
)
```

Purpose:

- Analyze revenue stability
- Support forecasting analysis



Figure 1.23: Forecasting KPI Cards

Revenue Trend Over 6 Months:

A **Line Chart** was implemented to visualize revenue trends over time and apply forecasting analysis.

Configuration:

- X-Axis: InvoiceDate
- Y-Axis: Total Revenue
- Forecasting Enabled
- Forecast Length: 6 Months
- Confidence Interval: 95%
- Forecast Units: Points

Purpose:

- Predict future revenue performance
- Identify seasonal trends and growth patterns
- Support strategic business planning

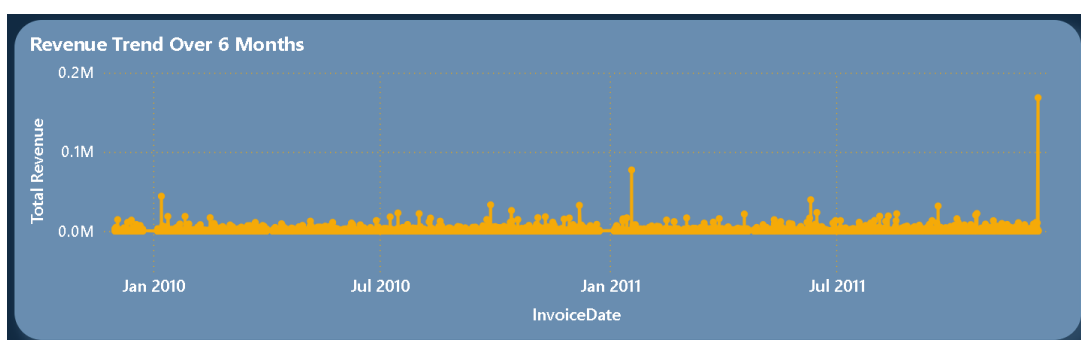


Figure 1.24: Forecasting Prediction Chart

Additional Calculated Columns

Is Cancelled A calculated column named **Is Cancelled** was created to classify invoices as either Cancelled or Completed.

Invoices beginning with the letter “C” were identified as cancelled transactions.

```
Is Cancelled =  
IF(  
    LEFT(FORMAT(FactSales[Invoice], ""), 1) = "C",  
    "Cancelled",  
    "Completed"  
)
```

Purpose:

- Support cancellation analysis
- Improve operational reporting accuracy

1.4.6 Conclusion

The implemented dashboard successfully transformed raw sales and customer data into interactive business intelligence insights using Power BI. Through the use of DAX measures, calculated columns, KPI cards, forecasting models, and interactive visualizations, the system provides valuable analytical capabilities for customer analysis, product performance evaluation, and future revenue prediction.

The dashboard enables decision-makers to monitor business performance efficiently, identify trends, improve customer retention, optimize product strategies, and support data-driven decision-making processes.

Chapter 2

Python

2.1 Data Preparation

2.1.1 Data Collection and Integration

- Import CSV files
- Verify schema consistency

2.1.2 Data Cleaning

The dataset required several preprocessing steps before analysis.

Handling Missing Values

```
dataset.dropna(inplace=True)
```

Removing Duplicate Rows

```
dataset.drop_duplicates(inplace=True)
```

Fixing Date Formats

```
dataset['InvoiceDate'] = pd.to_datetime(dataset['InvoiceDate'])
```

Removing Invalid Transactions

The following records were removed:

- Cancelled invoices
- Negative quantities
- Invalid prices

Removing Outliers

To make the dataset cleaner and more accurate, outliers were removed using the IQR (Interquartile Range) method. The IQR method calculates:

$$\text{IQR} = \text{Q3} - \text{Q1}$$

Values outside the normal range were considered outliers and removed from the dataset. This improved data quality and model accuracy.

```
Q1 = data['TotalPrice'].quantile(0.25)
Q3 = data['TotalPrice'].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

data = data[(data['total_order_price'] >= lower_bound) &
            (data['total_order_price'] <= upper_bound)]
```

2.1.3 Feature Engineering

Revenue Column

```
dataset['Total_order_price'] =
dataset['Price'] * dataset['Quantity']
```

Date Features

Additional date features were created:

- Year
- Month
- Day
- Hour

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

2.2 Dataset Split and Time-Based Validation

To improve the reliability of the analysis, the dataset was divided into two separate transactional periods:

- **Training Period (2009–2010):** Used to build RFM scores, define customer segments, and analyze purchasing behavior patterns.
- **Testing / Validation Period (2010–2011):** Used to validate customer segment stability, evaluate churn behavior, and compare purchasing activity over time.

This time-based validation approach allowed the project to evaluate whether customer classifications from the first period remained consistent in the following year.

2.3 Machine Learning Model

2.3.1 Model Objective

The objective of the machine learning model is to predict customer order value.

2.3.2 Model Selection

The Random Forest Regressor model was selected because:

- It handles nonlinear relationships
- It performs well on tabular data
- It reduces overfitting

2.3.3 Training Process (2009-2010)

```
features = ['Price', 'Quantity']
target = 'Total_order_price'

x_train, y_train = train[features], train[target]
x_test, y_test = test[features], test[target]
```

2.3.4 Model Evaluation

```
mae = mean_absolute_error(y_test, predictions)
```

The Mean Absolute Error (MAE) was used to evaluate model accuracy.

2009-2010

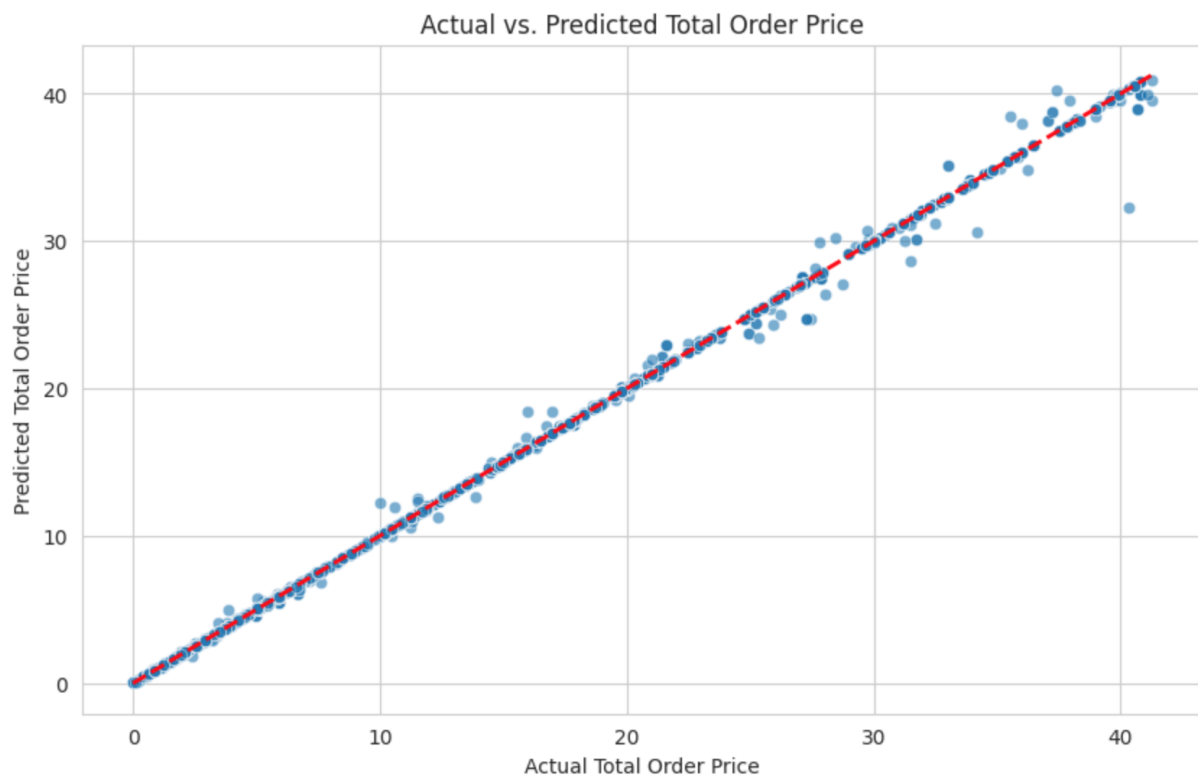


Figure 2.1: Model Accuracy Scatter Plot 2009-2010

The machine learning model achieved excellent predictive performance based on the evaluation metrics obtained. The Mean Absolute Error (MAE) of \$0.01 indicates that the model's predictions differ only slightly from the actual order values on average. Additionally, the Root Mean Squared Error (RMSE) of \$0.11 demonstrates that prediction errors remain very small even when larger deviations are considered.

2010-2011

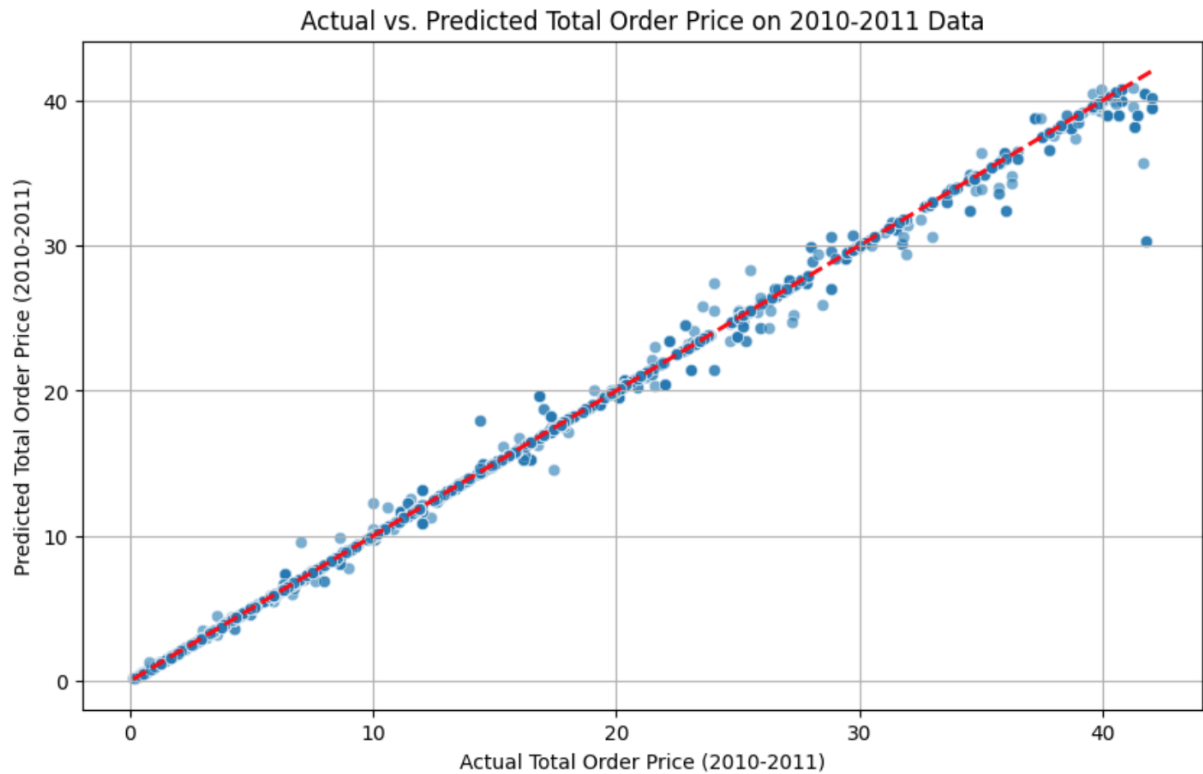


Figure 2.2: Model Accuracy Scatter Plot 2010-2011

The machine learning model achieved strong predictive performance when evaluated on the 2010–2011 validation dataset.

The Mean Absolute Error (MAE) of \$0.02 indicates that the model predictions were very close to the actual order values on average.

Additionally, the Root Mean Squared Error (RMSE) of \$0.21 demonstrates that prediction errors remained low even when larger deviations were considered.

Conclusion

Overall, the results demonstrate that the Random Forest Regressor model generalized effectively to future transactional data and can support forecasting and business decision-making processes.

2.4 RFM Analysis

2.4.1 RFM Metrics

Metric	Meaning
Recency	How recently a customer purchased
Frequency	How often a customer purchases
Monetary	How much money a customer spends

2.4.2 Customer Segmentation

Customers were segmented into:

- **Best Customers:** These are the most valuable customers, scoring high across all RFM metrics (Recency, Frequency, and Monetary).
- **Other Customers:** This is the largest customer group, representing customers who do not fit into the more specific high-value or at-risk categories.
- **Churned Customers:** Customers with low Recency and Frequency scores, indicating that they have not purchased recently and purchase infrequently.
- **Loyal Big Spenders:** Customers who purchase frequently and spend large amounts of money, although they may not be the most recent customers.
- **New Customers:** Recently acquired customers who may not yet have high Frequency or Monetary values.
- **Loyal Customers:** Customers who purchase frequently and consistently over time.
- **Big Spenders:** Customers with high Monetary scores who contribute significantly to overall revenue.
- **At-Risk Loyal Customers:** Previously loyal and frequent customers who have not made recent purchases, indicating a potential risk of churn.

2.4.3 RFM Segment Validation Across Time

To evaluate the stability of customer segments, customer behavior from the 2009–2010 period was compared against the 2010–2011 validation period.

Summary of RFM Segment Migration

- **Retained Best Customers:** 29 customers remained in the “Best Customers” segment across both periods.
- **Churn from High-Value Segments:**
 - 4 “Best Customers” became “Churned Customers” and 54 moved to “Other Customers”.

- 0 “Loyal Customers” became “Churned Customers” and 25 moved to “Other Customers”.
- 5 “Loyal Big Spenders” became “Churned Customers” and 52 moved to “Other Customers”.

- **Upgrades and Downgrades:**

- 11 customers from “Other Customers” upgraded to “Best Customers”.
- 1 customer from “New Customers” upgraded to “Best Customers”.
- 3 “Best Customers” became “At-Risk Loyal Customers”.

These findings demonstrate that customer purchasing behavior changes over time and highlight the importance of continuous customer monitoring and retention strategies.

2.4.4 RFM Analysis Conclusion

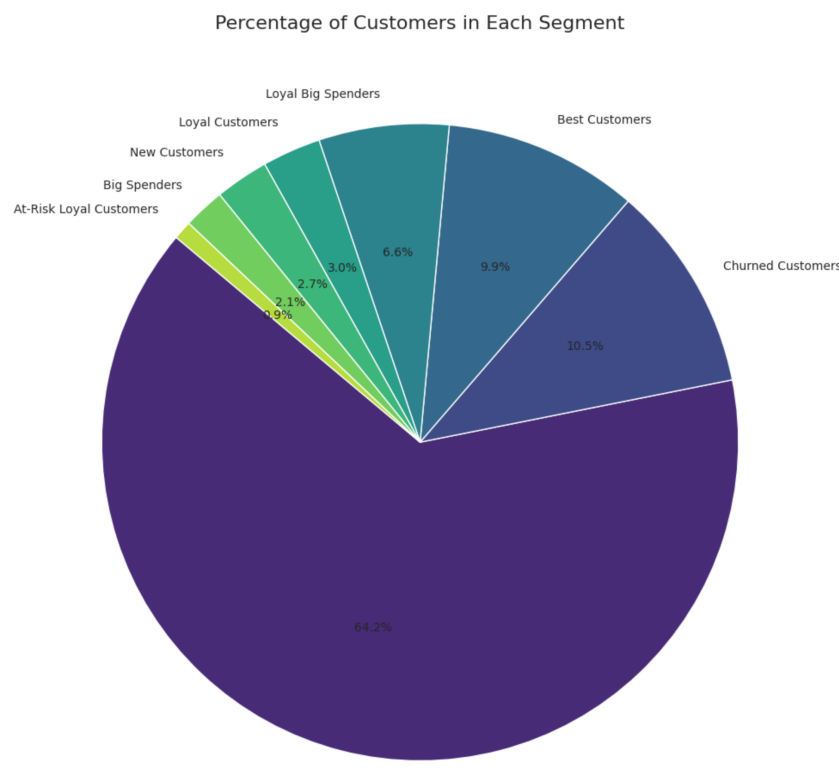


Figure 2.3: Customer Segmentation (2009-2010)

The RFM analysis provided valuable insights into customer purchasing behavior and customer value distribution.

The results show that the largest customer segment belongs to the “Other Customers” category, representing approximately 64.2% of all customers. This indicates that the majority of customers demonstrate moderate purchasing behavior and do not strongly belong to either high-value or at-risk groups.

The analysis also identified that 9.9% of customers are classified as “Best Customers,” while 6.6% are categorized as “Loyal Big Spenders.” These customers contribute significantly to overall revenue and should be prioritized for retention strategies and loyalty programs.

Additionally, 10.5% of customers were identified as “Churned Customers,” indicating a considerable number of inactive or disengaged customers. This highlights the need for re-engagement campaigns and personalized marketing efforts.

Smaller customer groups such as “New Customers,” “Loyal Customers,” and “Big Spenders” represent opportunities for growth through targeted promotions and customer relationship management strategies.

The “At-Risk Loyal Customers” segment, although relatively small, is particularly important because these customers were previously active and valuable but may soon churn if no retention actions are taken.

Using the 2010–2011 validation dataset, the analysis showed that several customers remained loyal over time, while others transitioned into lower-value or churned segments. Additionally, some customers improved their purchasing behavior and moved into higher-value customer groups, demonstrating that customer value changes dynamically over time.

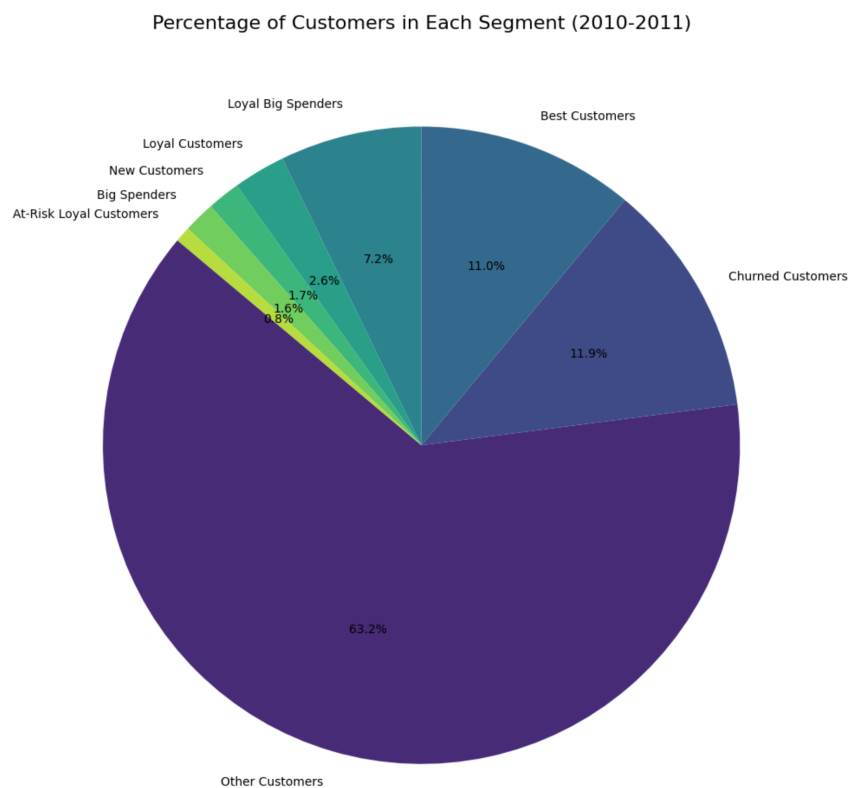


Figure 2.4: Customer Segmentation (2010-2011)

Overall, the RFM analysis demonstrates that customer segmentation can support data-driven marketing strategies, improve customer retention, and increase long-term business profitability.

2.4.5 Action Plans

- **Retain Best Customers:** Introduce loyalty rewards, exclusive discounts, and personalized offers for high-value customers to maintain customer satisfaction and encourage repeat purchases.
- **Re-engage Churned Customers:** Launch targeted email campaigns and promotional offers aimed at inactive customers to encourage them to return and make new purchases.
- **Strengthen Relationships with Loyal Big Spenders:** Provide premium services, early product access, and customized recommendations to maintain long-term customer loyalty.
- **Develop New Customers:** Create onboarding campaigns and personalized communication strategies to convert new customers into repeat buyers.
- **Reduce Customer Churn:** Monitor customer purchasing frequency and implement early-warning systems to identify customers showing declining engagement.
- **Increase Average Order Value:** Apply cross-selling and upselling strategies for customers with high purchasing potential.
- **Improve Marketing Personalization:** Use customer segmentation results to create tailored marketing campaigns based on customer behavior and spending patterns.

2.5 SMART Recommendations

Objective		SMART Goal
Customer Retention		Increase retention of Best Customers by 15% within the next 6 months through loyalty programs and personalized promotions.
Churn Reduction		Reduce the percentage of Churned Customers by 10% within 4 months using targeted re-engagement campaigns.
Sales Growth		Increase average customer spending by 12% within the next two quarters through upselling and cross-selling strategies.
Customer ment	Engage-	Improve repeat purchase rates among New Customers by 20% within 6 months through onboarding campaigns and follow-up marketing.

Marketing Efficiency	Implement segmented marketing campaigns for all major RFM customer groups within the next 3 months to improve campaign effectiveness.
Revenue Optimization	Increase revenue contribution from Loyal Big Spenders by 10% within 6 months through premium offers and personalized recommendations.

2.6 Challenges Faced

- Missing customer IDs
- Invalid transaction records
- Cancelled invoices
- Missing values after training